

EEF Meeting Minutes, Thursday March 16, 2023

Erlang Ecosystem Foundation, Minutes of a Regular Meeting of the Board of Directors Meeting #78, March 16, 2023

A regular meeting of the Erlang Ecosystem Foundation was held over group conference call (Zoom). The meeting began at 16:03 UTC.

Attendance

The following members of the board of directors were present (9/11): Francesco Cesarini, Maxim Fedorov, Ben Marx, Frédéric Trottier-Hébert, Bryan Paxton, Kenneth Lundin, Miriam Pena Villanueva (at beginning and end), Sebastian Strollo, and Alistair Woodman.

The following members of the board of directors were absent: Sophie DeBenedetto, Peer Stritzinger

Also present for the first item on the agenda were two guests: José Valim and Tristan Sloughter (invited to present two stipend proposals).

Agenda Items

1. Stipend Requests:

1. Erlang and Elixir APIs to access full TensorFlow Lite C/C++ APIs, from Ruomeng Xu (Cocoa) for \$12,000. The project was presented by José, who also answered questions to everyone's satisfaction.

RESOLVED, that the Erlang Ecosystem Foundation approve the stipend request from Ruomeng Xu for \$12,000 to implement an Erlang and Elixir API to TensorFlow Lite. Unanimously approved.

2. Implementation of Apache Arrow “Encapsulated Message Format” and “Apache Arrow IPC”, from Benjamin Philip for \$2,500. The project was presented by Tristan and José.

RESOLVED, that the Erlang Ecosystem Foundation approve the stipend request from Benjamin Philip for \$2,500 to implement the Apache Arrow Encapsulated Message Format and IPC. Unanimously approved.

3. Gazillion Funding (as Miriam was absent during this item in the agenda it was not presented or voted on)
2. Approval of minutes from previous meetings.

RESOLVED, that the Erlang Ecosystem Foundation approve the minutes as presented as representing a true and accurate record of the board meeting #77 held on March 2, 2023. Unanimously approved

3. Intermediate actions and activities: none
4. Treasury and membership update: Sebastian presented a quick summary of the current financial status, see table below. No membership update as Peer was absent.

Cash in Bank	\$	189,950
Pledged Sponsorship (not yet received)	\$	36,000
Pledged Stipends (not yet paid out)	\$	(34,950)
	\$	191,100

5. Unanimous consent votes: none scheduled
6. Board election status. Bryan and Ben reported: Dates moved out one week (due to mixup), Bryan is documenting process for next year. Friday is the last day for candidate submission. Hard to find candidates, but have three new candidates so far. Clarification: it is not a requirement to be a member to be a candidate.
7. Alistair provided an update on the EU Cyber Resiliency Act (CRA) and Product Liability Directive (PLD). Alistair will open a separate Slack channel for this and keep an eye on the development.
8. Paid assistant role: the meeting looked through the current status of the job description. Action for everyone (still): add your ideas to the document.
9. Budget planning 2023. Skipped.
10. Any other business:

1. Sebastian proposed that EEF registers with techsoup.org to receive non-profit discounts on many services (such as Slack, Notion, etc). None against, Sebastian will proceed.
 2. Ben brought up the fact that ElixirConf has announced that they will have a conference in Florida (the 1st week in September). Brief discussion around how the EEF best support inclusivity (given the new laws in the state of Florida). Alistair, Ben, and Fred to follow up and contact the organizers and ask if they have considered the implications of holding a conference in Florida.
 3. Bryan brought up the stipend approval, payment, follow up process: how can we offload, and make the process smoother.
 4. Bryan brought up sponsoring hex.pm, will be carried over to next meeting.
11. Alistair declared the meeting adjourned at 17:01 UTC.

Dated March 16, 2023

Secretary of the meeting: _____ Sebastian Strollo

Attachments:

1. Stipend request from Benjamin Philip regarding implementation of Apache Arrow Encapsulated Message Format and IPC (1 page)
2. Stipend request from Ruomeng Xu (Cocoa) regarding TensorFlow Lite integration (2 + 4 pages)

Stipend Proposal

Stipend Type : Development Work

First name : Benjamin

Last Name : Philip

Email Address : benjamin.philip495@gmail.com

City : Bengaluru

Region : Karnataka

Country : India

Intended product of project :

This project intends to create a working Erlang implementation of Apache Arrow in order to serialize and deserialize Erlang terms into (and from) Apache Arrow's Encapsulated Message Format, Apache Arrow IPC (both streaming and file) and Apache Flight. In the future, if there is demand, support for Compute functions and other features similar to that in PyArrow may be added through a separate library.

Purpose

This project will benefit both the Observability Working Group and the Machine Learning Working Group. The Observability WG needs a working Arrow implementation to adhere to a proposal to the OpenTelemetry Protocol: <https://github.com/open-telemetry/oteps#171>. Though there aren't any plans to use this project in the ML WG, it can be used read and write Nx tensors from and to Arrow, to develop a new backend for Explorer, and a lot more.

Development Plan

I, Benjamin Philip, plan to work on this project from March 22nd till June 1st fulltime. Support for the basic encapsulated format will be worked on from March till Mid or End of April. This will make the project in a usable state for Erlang's telemetry implementation. I will be working on adding IPC and Flight support from May till June. After this I will release version v0.1.0. From June onwards, this project will only be maintained and not be actively developed. Support for the afore mentioned compute functions will only be added later if it benefits the ML WG or any other party.

Dissemination : This is not a very "public facing project", and will likely not be used by people directly but as a dependency to telemetry, nx, explorer and similar. Therefore, dissemination will likely be limited to posts on the Elixir and Erlang Forums, Blog posts etc. Members of the Observability and ML WGs are already aware of this project.

Delivery Method : I will be supervised primarily by Tristan Sloughter of the Observability Working Group, and also by José Valim of the Machine Learning Working Group. As mentioned in the Development Plan, I will be working on this project fulltime.

Stipend Amount (USD) : 2500

Stipend Proposal

Stipend Type : Development Work

First name : Ruomeng

Last Name : Xu

Nick Name : Cocoa

Email Address : i@uwucocoa.moe

City : Glasgow

Country : United Kingdom

Twitter : @_uwu_cocoa

Linkedin : <https://www.linkedin.com/in/cocoa-xu/>

Website : <https://uwucocoa.moe>

Intended product of project :

A compact machine-learning inference library that provides Erlang and Elixir APIs to access full TensorFlow Lite C/C++ APIs, good documentation, and 5 examples (in the form of both Livebook and pure Erlang code) covering image classification, object detection, segmentation, super-resolution, and audio classification.

Purpose

This project aims to create a compact machine-learning inference library powered by TensorFlow Lite. - Create a compact machine-learning library. As aforementioned, TensorFlow Lite has a small footprint which makes it a great option for users who want a compact machine-learning inference library. - Make machine-learning inference tasks on Erlang Ecosystems more accessible on resource-constrained systems.

Development Plan

(Please see the attached file for better formatting and more details) i) Porting TensorFlow Lite APIs. In this stage, I'll port TensorFlow Lite APIs and organize them in corresponding Erlang and Elixir modules. Existing ones in Elixir will also be ported into Erlang. Proper inline documentation will be added while porting each API. This stage is estimated to take about 3 months of time. Including writing unit tests in Elixir along the way. ii) Provide an Elixir wrapper, including one that interacts with Nx (Numerical Elixir), using structs and implementing Nx.LazyContainer for the data in and out. This stage is estimated to take about 1 month of time. iii) Thorough testing. As this library will be implemented in NIF and considering its use cases, stability is a key factor. At this stage, I intend to test this library on mainstream platforms: • x86_64-linux-gnu • aarch64-linux-gnu • x86_64-apple-darwin • aarch64-apple-darwin The goal is to cover at least 80% of the code. This stage is estimated to take about 1 month of time. We intend to provide a single package for both Erlang and Elixir, and we will make sure that the installation works via rebar3 for Erlang developers. Meanwhile, I intend to rename the existing GitHub repo from "tflite_elixir" to "tflite_beam". The name combines "TFLite" from TensorFlow Lite and "Beam" from Erlang's BEAM virtual machine, which is a fitting reference to the technology stack that the library supports. Of our goals, only the

second goal is specific to Elixir. The remaining of our efforts should benefit the Erlang Ecosystem as a whole.

Dissemination : 1. Blog posts. The following series of blog posts on what this library can do and how to use it: (i) How to install and setup everything, and ends with a demo showing an image classification task on both CPU and Coral edge TPU. (ii) How to integrate with Nerves, i.e., using this library on embedded devices (e.g., Raspberry Pi). (iii) Additional five short blog posts (one for each example). These blog posts will include screenshots or screen recordings so that viewers can see the whole process before they install this library. 2. Conference talk If possible and accepted by a conference, I intend to show how to load TFLite models, and get the predicted results. These demos will include the official ones on TensorFlow Lite's website. 3. Social Network I'll post the latest progress on my Twitter and Mastodon accounts while doing the development work with commonly used hashtags, e.g., #webeamtogether and #myelixirstatus.

Delivery Method : This project is and will stay open-source on GitHub for everyone to access, use and contribute. We also intend to publish the packages to hex.pm, whenever it is ready to ship a newer version.

Stipend Amount (USD) : 12000

Apply for the EEF stipends - TFLiteBEAM: TensorFlow Lite Bindings

Cocoa

Application

1. Stipend Type - Development Work

TensorFlow Lite is a lightweight version of the popular open-source machine learning framework TensorFlow. It is optimized for on-device machine learning, and it achieves this goal by addressing 5 key constraints:

- latency: there's no round-trip to a server.
- privacy: no personal data leaves the device.
- connectivity: internet connectivity is not required.
- size: reduced model and binary size.
- power consumption: efficient inference and a lack of network connections.

Therefore, TensorFlow Lite is widely used in applications such as mobile image and speech recognition, natural language processing, and object detection. With TensorFlow Lite, developers can create intelligent applications that run on low-power, resource-constrained devices, bringing the power of machine learning to the edge.

Current status. *tflite_elixir*¹ is a work in progress FOSS (Free Open-Source Software) licensed under Apache 2.0 on GitHub. It aims to create an Erlang/Elixir library that includes both high-level and low-level APIs. High-level APIs will be designed as one or more modules that take the necessary parameters and return the inferred results to the caller. Low-level APIs include all TensorFlow Lite C/C++ APIs, which allow users to have finer control over this library and create their own high-level modules.

¹ https://github.com/cocoa-xu/tflite_elixir

Some initial findings of the project have the following advantages,

1. **Small footprint.** It could be a great option for users who want a compact machine-learning inference library. Initial releases of this library were less than **10 MBs** whereas *Torchx*/*EXLA* usually takes at least 100-200MB.
2. **Works on various systems.** Initial releases of this library already have precompiled versions for 64-bit RISC-V CPUs while compiling/cross-compiling both *Torchx* and *EXLA* for RISC-V was unsuccessful (as of the time of writing).

2. *Intended product of the project*

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3. *Purpose*

This project aims to create a compact machine-learning inference library powered by TensorFlow Lite.

- Create a compact machine-learning library. As aforementioned, TensorFlow Lite has a small footprint which makes it a great option for users who want a compact machine-learning inference library.
- Make machine-learning inference tasks on Erlang Ecosystems more accessible on resource-constrained systems.

4. *Development Plan*

i *Porting TensorFlow Lite APIs.*

In this stage I'll port TensorFlow Lite APIs and organize them in corresponding Erlang and Elixir modules. Existing ones in Elixir will also be ported into Erlang. Proper inline documentation will be added while porting each API.

This stage is estimated to take about 3 months of time. Including writing unit tests in Elixir along the way.

- ii Provide an Elixir wrapper, including one that interacts with Nx (Numerical Elixir), using structs and implementing *Nx.LazyContainer* for the data in and out.

This stage is estimated to take about 1 month of time.

- iii Thorough testing. As this library will be implemented in NIF and considering its use cases, stability is a key factor. In this stage I intend to test this library on mainstream platforms:

- x86_64-linux-gnu
- aarch64-linux-gnu
- x86_64-apple-darwin
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The goal is to cover at least 80% of the code.

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Of our goals, only the second goal is specific to Elixir. The remaining of our efforts should benefit the Erlang Ecosystem as a whole.

5. Dissemination

- Blog posts. The following series of blog posts on what this library can do and how to use it:
 - (i) How to install and setup everything, and ends with a demo showing an image classification task on both CPU and Coral edge TPU.

(ii) How to integrate with Nerves, i.e., using this library on embedded devices (e.g., Raspberry Pi).

(iii) Additional five short blog posts (one for each example).

These blog posts will include screenshots or screen recordings so that viewers can see the whole process before they install this library.

- Conference talk

If possible and accepted by a conference, I intend to show how to load TFLite models, and get the predicted results. These demos will include the official ones on the TensorFlow Lite's website.

- Social Network

I'll post the latest progress on my Twitter and Mastodon accounts while doing the development work with commonly used hashtags, e.g., *#webeamtogether* and *#myelixirstatus*.

6. *Delivery*

This project is and will stay open-source on GitHub for everyone to access, use and contribute. We also intend to publish the packages to *hex.pm*, whenever it is ready to ship a newer version.

7. *Amount*

12,000 USD for 5 months. This will cover the applicant's salary for 5 months to further develop this TensorFlow Lite Erlang/Elixir library as part-time work alongside the applicant's full-time PhD studies (Oct. 2021 - Oct. 2024).